EARTHPRESS · STEM CHECK

EarthPress STEM check

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-25	Amish Chadha	First STEM check (kit 1.8 pipeline, run during the kit 1.9.1 upgrade) for the current design (option A) and options B and C; S9 and S10 failed on the 35 mm C45 toggle pins and were closed by fix F1 under the standing instruction of 2026-10-23; independent review recorded

EarthPress STEM check

Verdict: compacting moist, cement-stabilized soil into a 290 x 140 x 90 mm block at 2 MPa with a two-person lever and toggle press sits within known physics, chemistry and biology, and no row fails. Two rows failed when first run, both on the toggle pins: the 35 mm C45 base pin yielded in bending at the nominal 81.2 kN (388 MPa against 370 MPa) and the 20 mm base pin lugs exceeded the bush bearing allowable at the 173 kN design load. Both were closed by fix F1 (40 mm 42CrMo4 +QT pins, 28 mm lugs; EPR-DEC-001), decided on 2026-10-25 under Amish's standing instruction of 2026-10-23. Of 35 rows, 23 are within known science and 12 are plausible, to confirm, so `stem_check` is **open**: for option A the soil compaction law (S6), the block strength at 2 MPa (S2), the lime dose (S18), the operator force and eject force (S5, S14), the fill scale (S8) and the team lift (S21) need measurements or assessment on partner soils, which are TRL 4 work and on hold. Method: STANDARDS section 20. Calculations: `docs/04-calcs/stem_check.py`, which takes option A's numbers from EPR-CAL-001 (`docs/04-calcs/sizing.py`).

Claims checked

Table 1. Claims, governing science and verdicts (generated from `docs/04-calcs/stem_results.csv`; rows S are option A, the current design; B and C are the other concept options).

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
S1	The press needs 81.2 kN on the 290 x 140 mm block face for 2 MPa	Mathematics	Units and dimensional analysis	F = p A	claimed 81.2 kN, estimate 81.2 kN (ratio 1.00)	Within known science	EPR-CAL-001
S2	Compacting a 5 % cement mix to 2 MPa gives blocks that can meet the R3 strength class with suitable soil	Physics	Mechanics and statics	Auroville Earth Institute: CSEB pressed at 2 to 4 MPa with about 5 % cement; the Auram press 3000 works at 2.7 to 5.3 MPa	2 MPa is the low end of practice; strength depends on soil, mix and curing and is measured in block tests (TRL 4, on hold)	Plausible, to confirm at TRL 3	Open
S3	The lever and toggle give a force ratio of 173:1 at the crank stop	Physics	Mechanics and statics	Toggle 1 / (2 tan 6 deg) times grip radius over the connecting link's arm (virtual work)	claimed 173 :1, estimate 201 :1 (ratio 0.86)	Within known science	EPR-CAL-001
S4	Two people on the 1.71 m grip path supply the compaction work of the softer design soil	Physics	Mechanics and statics	Conservation of energy or mass	in 1710 J, out 705 J	Within known science	EPR-CAL-001
S5	Each of two operators pulls no more than 500 N at the peak, with the softer design soil	Biology	Ergonomics	Peak grip force from the force ratio and the soil law (CAL section 4b); project limit 500 N per operator (R5, PR-005); ISO 11228-2 screening to assess	405 N against 500 N, but it rests on the assumed soil law (S6) and on equal shares between the two operators; measure on partner soils	Plausible, to confirm at TRL 3	Open
S6	Soil pressure in the mold rises tenfold over the last 14 mm of the	Physics	Mechanics and statics	Assumed exponential compaction law; no measured curve for the partner's soils	Measure pressure against travel on partner soils (compaction test); S4, S5 and the fill window rest on it	Plausible, to confirm at TRL 3	Open

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	stroke (10 to 20 mm)						
S7	Every load-path member stays below yield at the 173 kN design load (1,000 N at the grip at the stop)	Physics	Strength of materials	Bending, shear, bearing, net section and buckling against yield of S235/ S250 plate, A53 pipe and 42CrMo4 +QT pins (CAL Table 5)	1 of yield <= 1.29 of yield	Within known science	EPR-CAL-001
S8	A 10 kg scale with 50 g divisions can hold the fill to its window of 0 to +206 g	Engineering and technology	Sensing resolution	Scale division against half the fill window (CAL section 5)	50 g division against a half-width of 103 g for the base soil, but the softer design soil has a window of about 0 to +153 g (half-width 76 g), a division is not the scale's accuracy (a low-cost 10 kg scale is about 50 to 100 g) and a 1 % moisture swing moves the dry mass by about 70 g: confirm scale accuracy and moisture control	Plausible, to confirm at TRL 3	Open
S9	The 40 mm 42CrMo4 +QT toggle pins stay below yield in bending at the 173 kN design load	Physics	Strength of materials	Pin in a clevis: link force per side times the distance between bearing centres (base pin 36 mm arm), M / (pi d^3 / 32) against Rp0.2 750 MPa for 16 < d <= 40 mm; the 35 mm C45 pins of CAL v0.6 failed (388 MPa against 370 MPa at the nominal load, 826 MPa at 173 kN)	498 MPa <= 750 MPa	Within known science	EPR-DEC-001 F1 (2026-10-25, standing instruction)

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
S10	The base pin lugs carry the design load within the bush bearing allowable	Physics	Strength of materials	Bearing p = F / (2 t d) on 2 x 28 mm lugs at 40 mm, allowable 100 MPa for hardened greased bushes (assumed); the 20 mm lugs on 35 mm reached 124 MPa	78 MPa <= 100 MPa	Within known science	EPR-DEC-001 F1 (2026-10-25, standing instruction)
S11	Welded members last 50,000 blocks in fatigue	Physics	Strength of materials	Nominal stress range per block against a FAT 71 detail at 50,000 cycles (EN 1993-1-9 method)	89 MPa <= 243 MPa	Within known science	EPR-CAL-001
S12	The turned toggle pins last 50,000 blocks in bending fatigue	Physics	Strength of materials	Pulsating range against 0.5 Rm x 0.7 for surface and size (Rm 1,000 MPa minimum, 42CrMo4 +QT; first-order)	234 MPa <= 350 MPa	Within known science	EPR-CAL-001
S13	Released at the stop, the lever kicks back with about 47 J	Physics	Mechanics and statics	Strain energy 0.5 F d with 1.0 mm soil rebound (assumed) and the column stretch	claimed 47 J, estimate 47.1 J (ratio 1.00)	Within known science	EPR-CAL-001
S14	One person ejects the block with 470 N or less on the eject lever	Physics	Mechanics and statics	Wall friction 0.1 MPa residual lateral pressure x 0.6 x wall area, plus piston and block weight, over the 10.3:1 eject ratio (assumed residual pressure)	469 N against 500 N, a margin of about 6 % that rests on the assumed 0.1 MPa residual pressure (at 0.2 MPa the force would be about 930 N): measure on partner soils	Plausible, to confirm at TRL 3	Open
S15	A crew of four makes about 296 blocks in an 8 h day	Mathematics	Orders of magnitude	Makiga manual press, 290 x 140 mm blocks, about 500 a day with a	claimed 296 blocks, estimate 500 blocks (ratio 0.59)	Within known science	Open

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
				crew of four (Engineering for Change); EarthPress cycle 85 s assumed (CAL section 9)			
S16	Curing blocks bind about 0.23 kg of water per kg of cement	Chemistry	Physical chemistry	Non-evaporable water of fully hydrated Portland cement about 0.23 g/g (Powers and Brownyard; Neville, Properties of Concrete)	Used only for the cured mass in the flow figure; 712 kg per 100 blocks	Within known science	EPR-CAL-001
S17	CSEB walling embodies 25 % or less of the CO2 of fired brick of equal volume	Chemistry	Environmental and atmospheric chemistry	Auroville Earth Institute factors: 49 kg CO2/m3 for 5 % CSEB, 643 kg CO2/m3 for locally fired brick	7.6 % <= 25 %	Within known science	EPR-CAL-001
S18	Lime at about 8 % stabilizes clay-rich soils, with a longer curing period than cement	Chemistry	Inorganic chemistry	Lime and clay react slowly (cation exchange, then pozzolanic calcium silicate and aluminate hydrates); needs enough clay and moisture	Starting value only; wet-strength tests on partner soils decide the dose and the curing time (TRL 4, on hold)	Plausible, to confirm at TRL 3	Open
S19	Sieving dry soil raises respirable crystalline silica, so damp sieving and an FFP2 or N95 respirator are specified	Biology	Toxicology	Crystalline silica is carcinogenic to humans (IARC Group 1, Monograph 100C); OSHA PEL 50 ug/m3 over 8 h (29 CFR 1926.1153)	Hazard and controls stated in PRC, BLD and RSK; no exposure claim is made	Within known science	EPR-RSK-001
S20	Wet cement can burn skin and eyes, so gloves and eye protection are specified	Biology	Toxicology	Portland cement safety data sheets: causes skin irritation and serious eye damage (hazard statements H315,	Hazard and controls stated in PRC, BLD and RSK	Within known science	EPR-RSK-001

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
				H318); wet mix is strongly alkaline			
S21	Two people can lift the heaviest piece of the press	Biology	Ergonomics	ISO 11228-1 reference mass 25 kg per person for occasional lifts; two people share the piece	22.2 kg per person if the load halves exactly, but team lifting usually reduces each person's share of the capacity and the 25 kg figure is for ideal conditions: to assess; lift with a trolley, a hoist or four people	Plausible, to confirm at TRL 3	Open
S22	The grip stays at 0.8 m or higher through the stroke, with the peak force at 1.0 m or higher	Biology	Ergonomics	Lever kinematics (CAL section 3); R5 band	800 mm <= 863 mm	Within known science	EPR-CAL-001
S23	The mold walls bulge far less than the 2 mm block tolerance at 2 MPa	Physics	Strength of materials	Fixed-fixed wall strip with the belt, lateral pressure 0.5 x vertical (assumed K0)	0.078 mm <= 2 mm	Within known science	EPR-CAL-001
B1	Option B: two people on a 1.2 m capstan turn the screw at the 81 kN pressing force within 500 N each	Physics	Mechanics and statics	Power screw torque T = F d2 / 2 tan(lambda + phi'), lambda 7.0 deg, phi' 8.8 deg (mu 0.15), T 430 N m; grips 0.53 m from the axis	406 N <= 500 N	Within known science	Open
B2	Option B: the two-start screw is self-locking, so it holds the load without a pawl	Physics	Mechanics and statics	Self-locking when the lead angle is below the friction angle	lambda 7.0 deg against phi' 8.8 deg at mu 0.15 but 5.9 deg at mu 0.10 (well greased): not self-locking when greased; a brake	Plausible, to confirm at TRL 3	Open

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
					or a single-start screw would be needed		
B3	Option B: a 90 mm bronze nut carries the pressing force within its bearing allowable	Physics	Strength of materials	Thread bearing p = F / (pi d2 H1 n); hand press, steel screw on bronze nut, 17 to 24 MPa (Shigley, Table 8-4)	15.7 MPa <= 17 MPa	Plausible, to confirm at TRL 3	Open
B4	Option B: the screw's root section stays below yield at the largest force two people can make	Physics	Strength of materials	Root diameter 32 mm; two people at 500 N on 0.53 m give the largest force; C45 yield 340 MPa (first-order)	128 MPa <= 340 MPa	Within known science	Open
B5	Option B: a crew of four makes about 225 blocks a day	Mathematics	Orders of magnitude	Makiga manual press about 500 a day (Engineering for Change); option B cycle 111 s with 14 unloaded turns each way (227 a day)	claimed 225 blocks, estimate 500 blocks (ratio 0.45)	Within known science	Open
B6	Option B: the operators supply the pressing work through a screw of about 43 % efficiency	Physics	Mechanics and statics	Conservation of energy or mass	in 1.432e+04 J, out 1633 J	Within known science	Open
C1	Option C: about 25 blows of a 20 kg weight falling 450 mm give standard Proctor compaction energy	Physics	Mechanics and statics	ASTM D698 standard effort 592.7 kJ/m3 over the 3.65 L block; blow energy m g h	claimed 25 blows, estimate 24.5 blows (ratio 1.02)	Within known science	Open

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
C2	Option C: impact compaction at standard Proctor effort reaches the density of 2 MPa static pressing	Physics	Mechanics and statics	Impact and static compaction give different density profiles; impact from the top densifies the top layer most	No source equates the two for CSEB; test blocks needed	Plausible, to confirm at TRL 3	Open
C3	Option C: each of two people pulls under 500 N on the rope, even when lifting the plunger for filling	Biology	Ergonomics	Weight, rod and plunger about 40 kg, shared by two over the head pulley	196 N <= 500 N	Within known science	Open
C4	Option C: the operators' sustained power stays within human capacity	Biology	Human physiology and biomechanics	25 blows of 88 J in 100 s, shared by two; sustained adult output about 75 W	11 W <= 75 W	Within known science	Open
C5	Option C: a crew of four makes about 160 blocks a day	Mathematics	Orders of magnitude	Makiga manual press about 500 a day (Engineering for Change); option C cycle 160 s (25 blows at 15 a minute plus 60 s)	claimed 160 blocks, estimate 500 blocks (ratio 0.32)	Plausible, to confirm at TRL 3	Open
C6	Option C: the mold, plunger and floor stop survive the impact peak force	Physics	Strength of materials	Impact force depends on the soil's stiffness at the end of compaction, not yet known	To be estimated from a measured soil stiffness (S6)	Plausible, to confirm at TRL 3	Open

Verdicts: **Within known science**, **Plausible, to confirm at TRL 3**, **Fails, fix proposed**, **Fails**.

Rows closed by a fix

Row	First run (CAL v0.6 design)	Fix	Rerun
S9	Fails, fix proposed: 35 mm C45 base pin in bending, 388 MPa against 370 MPa at 81.2 kN and 826 MPa at 173 kN (arm 40 mm from the link to the lug); knee and upper pins 661 and 641 MPa at 173 kN	F1: 40 mm 42CrMo4 +QT pins (Rp0.2 750 MPa), lugs grown to 28 mm so the arm falls to 36 mm (EPR-DEC-001, 2026-10-25, standing instruction)	Within known science: 498 MPa (base), 463 MPa (knee, with the connecting link), 429 MPa (upper) against 750 MPa at 173 kN
S10	Fails, fix proposed: 2 x 20 mm lugs on a 35 mm pin, 124 MPa against 100 MPa at 173 kN	F1: 2 x 28 mm lugs on the 40 mm bush bore	Within known science: 78 MPa

The fix keeps the working principle, the linkage geometry and the pitch; it is not a decision for Amish under the fix rule, but he can reverse it. It adds about 1.2 kg and about USD 19 (estimate), which takes the press to 203.8 kg against the 200 kg R7 total (decision D2, proposed, awaiting Amish) and the BOM to USD 542.

Fields not relevant

- Physics, electromagnetism and circuits, optics and radiation, nuclear and ionising radiation: no electrical parts, light or radiation sources; the press is hand powered (PR-015).
- Physics, thermodynamics and heat transfer: no heat process; cement hydration heat in a 90 mm block cured in shade is negligible for the press.
- Biology, microbiology, ecology and agronomy, epidemiology: no living material is used; soil with organic matter is screened out by the field tests (PR-009). Toxicology (S19, S20) and ergonomics (S5, S21, S22) are covered.
- Engineering and technology, power and link budgets, control: no powered or controlled parts.

Concept options

- Option A, lever and toggle press (the current design): rows S1 to S23; within known science after fix F1; open on S2, S6 and S18 (partner soil data).
- Option B, power screw press: within known science on force, thread bearing, screw strength, energy and output (B1, B3 to B6); plausible on self-locking (B2: the lead angle of a two-start Tr 40 x 14 screw is 7.0 deg against a friction angle of 8.8 deg at a friction coefficient of 0.15 but 5.9 deg at 0.10, so a greased screw may run back and would need a brake or a single-start screw) and on the nut, which is within 8 % of its bearing allowable (B3).
- Option C, rope-lifted drop rammer: within known science on blow count, rope pull and sustained power (C1, C3, C4); plausible on whether impact compaction reaches the density of 2 MPa static pressing (C2), on output, about 160 blocks a day, just under a third of the benchmark (C5), and on the impact force in the mold (C6).

No option has a Fails row, so none is dropped.

Independent review

The reviewer (claude-opus-5-5, run headless, shown only the STEM check, CAL, PRS and the two scripts) found nothing blocking and recomputed S1, S9, S10 (arithmetic), S15 to S16, S19, S20, S22, S23, B1 to B6 and C1, C3, C4 as holding. Findings and what was done under the fix rule:

- **S5 (operator force), S8 (scale), S14 (eject force), S21 (team lift): downgraded to Plausible, to confirm.** S5 rests on the assumed soil law (S6) and equal shares; S8 used the base soil's half-width (103 g) while the design soil's window is about 76 g and a 50 g division is not scale accuracy; S14 has a 6 % margin on an assumed 0.1 MPa residual pressure (930 N at 0.2 MPa); S21 assumed that two people halve the load exactly. The script and Table 1 now show these as plausible.
- **S3 (force ratio):** the closed-form estimate (201:1, symmetric knee) is 14 % above the 173:1 of CAL, which is the numerical derivative of the modelled linkage kinematics (the lever pivot, link and piston at their real positions, not a symmetric knee). The model is taken as authoritative and the closed form as an upper bound. Not changed; the design load stays at 173 kN. If the real ratio were 201:1 the lugs would reach about 91 MPa and the lowest factor about 1.11, so the linkage geometry should be re-checked in the TRL 4 prototype.
- **Design load basis (S7, S9, S10):** 1,000 N at the grip is two operators at their normal pull; two people hanging body weight (about 1,500 N) would give about 260 kN and exceed the lug allowable. The crank stop and end pawl are not separately checked. Proposed to Amish: either state the abuse case as 1,000 N and add a grip stop that limits the load, or raise the lug and pin sizes again. Not changed because it changes the stated design load and scope (the design load question).
- **S7 yield:** S235 plate is 235 MPa (225 MPa for 16 to 40 mm), not the 250 MPa used. The lowest factor of 1.29 is the base pin lug bearing, not a plate, so the headline holds; the mold wall factor falls from 1.31 to about 1.18. Noted, to correct in CAL v0.8.
- **S11, S12 (fatigue):** no partial factor, nominal load rather than the top of the fill window, no stress concentration at the grease hole, and the lug fillet welds are not checked (FAT 36 to 45). Margins are thin; noted, still "within known science" at first order, to be re-checked with the prototype.
- **S13 (kickback):** not independent of CAL; at the top of the fill window it would be about 80 J. The end pawl takes it either way.
- **S17 (carbon):** the cement count gives about 11 % rather than 7.6 %; the text now says so and R11 is still met.
- **Documents:** CAL has two tables numbered "5b" (to renumber in the next CAL revision).

Sources

- Auroville Earth Institute, "Compressed Stabilised Earth Block," https://www.earth-auroville.com/compressed_stabilised_earth_block_en.php, accessed 2026-10-25 (compaction 2 to 4 MPa, about 5 % cement; 49 and 643 kg CO2/m3; S2, S17).
- Auroville Earth Institute, "Specifications of the Press 3000," https://www.earth-auroville.com/specifications_of_the_press_3000_en.php, accessed 2026-10-25 (2.7 to 5.3 MPa, 150 kN; S2).
- Engineering for Change, "Makiga Stabilized Soil Block Press," updated 2024-03-04, https://www.engineeringforchange.org/solutions/product/makiga-stabilized-soil-block-press/, accessed 2026-10-25 (about 500 blocks a day with four workers; S15, B5, C5).
- Swiss Steel Group, "42CrMoS4" data sheet (Rp0.2 750 MPa for 16 < d <= 40 mm, +QT), https://swisssteel-group.com/content-media/documents/Data-Sheets/Engineering-Steel/42CrMoS4_en.pdf, accessed 2026-10-25 (S9, S12).

- OSHA, "Silica, Crystalline: Construction," https://osha.gov/silica-crystalline/construction-info, accessed 2026-10-25 (PEL 50 ug/m3, 29 CFR 1926.1153; S19).
- IARC, *Arsenic, Metals, Fibres, and Dusts*, IARC Monographs 100C (Lyon: IARC, 2012) (crystalline silica, Group 1; S19).
- ASTM D698, Standard Test Methods for Laboratory Compaction Characteristics of Soil Using Standard Effort (592.7 kN m/m3; C1).
- R. G. Budynas and J. K. Nisbett, *Shigley's Mechanical Engineering Design*, power screws and thread bearing pressure (Table 8-4) (B1 to B4).
- A. M. Neville, *Properties of Concrete*, non-evaporable water of hydrated cement (S16).
- ISO 11228-1 (lifting, reference mass 25 kg; S21) and ISO 11228-2 (pushing and pulling; S5, to assess).
- EN 1993-1-9, fatigue detail categories (FAT 71; S11).
- EPR-CAL-001 (`docs/04-calcs/01-sizing.md` , `sizing.py`) for every option A number.